

ALIYU ABUBAKAR TUKUR

Full-Stack Developer · Backend Engineer · AI & Automation

Abuja, Nigeria | +234 810 359 1545 | aliyuatk@gmail.com | aleeyutk.github.io | github.com/aleeyutk

PROFESSIONAL SUMMARY

Full-Stack Software Developer with a B.Tech in Computer Science and hands-on experience designing and building web applications, business management systems, REST APIs, secure backend services, and data-driven applications. Strong foundation in Java and Spring Boot, with practical experience across React, TypeScript, PostgreSQL, authentication and authorisation, database design, API integrations, Docker, and application deployment. Experienced in developing systems around real-world business workflows including role-based access control, multi-step approvals, transactional processes, analytics dashboards, audit trails, and external API integrations. Currently expanding into AI-powered applications, intelligent agent workflows, and automation.

TECHNICAL SKILLS

Languages	Java · TypeScript · JavaScript · Python · SQL
Backend & APIs	Spring Boot · Spring Security · Spring Data JPA · REST APIs · Node.js / Express · OpenAPI / Swagger
Frontend	React · TypeScript · Tailwind CSS · Vite · HTML · CSS
Databases	PostgreSQL · MySQL · H2 · Relational Database Design · SQL
DevOps & Tools	Docker · Docker Compose · Liquibase · Git · GitHub · Maven
Security & Architecture	JWT Authentication · Role-Based Access Control (RBAC) · API Security · Database Design · Audit Logging
Data & Integration	Chart.js · REST API Integration · External API Integration
Focus Areas	Artificial Intelligence · AI Agents · Full-Stack Development · Backend Engineering · DevOps · Secure Systems · Business Automation

PROFESSIONAL EXPERIENCE

Backend Developer Intern · FlexiSaf Edusoft Ltd

Jan 2021 – Dec 2021

- Assisted in developing and maintaining backend application logic using standard enterprise practices.
- Worked with databases and SQL queries to support application functionality, performance, and data management.
- Collaborated with cross-functional development teams to implement and deliver reliable software solutions.
- Gained practical experience in backend development workflows, debugging, problem-solving, and software delivery.

SELECTED PROJECTS

Hexa Operations — Restaurant Operations & Management Platform

React · TypeScript · Django · Expo · PostgreSQL

- Designed and developed a comprehensive restaurant operations platform focused on real-world business workflows and operational management.
- Built full-stack functionality spanning customer-facing experiences, restaurant operations, ordering workflows, and administrative processes.
- Developed responsive interfaces using React and TypeScript alongside mobile functionality using Expo.
- Implemented backend services and business logic using Django and REST-based application architecture.
- Integrated payment and real-time communication capabilities to support practical restaurant workflows.

- Structured the application as a maintainable multi-application workspace to support web and mobile clients.

Enterprise Mining Management System — Enterprise Workflow & Operations Platform

Java Spring Boot · React TypeScript · PostgreSQL · Docker

- Built a full-stack enterprise management system with JWT authentication and role-based access control supporting three user roles.
- Developed 25+ RESTful API endpoints using Spring Boot, Spring Security, and Spring Data JPA.
- Designed interactive analytics dashboards using Chart.js with 10+ data visualisations.
- Implemented automated business workflows covering mineral intake, invoicing, approval processes, and payment tracking.
- Designed a normalised PostgreSQL database and managed schema changes using Liquibase.
- Built a responsive React TypeScript frontend with form validation and dynamic real-time calculations.
- Developed an audit trail and asynchronous logging system with advanced search functionality.
- Containerised the application with Docker for reproducible deployment and environment management.

Country & Currency API — REST API & External Data Integration

Java 17 · Spring Boot 3 · H2 · Docker

- Developed a Spring Boot REST API that aggregates country and exchange-rate data from external APIs.
- Implemented country-data retrieval, caching, filtering, sorting, country lookup, API status monitoring, and data refresh functionality.
- Added GDP estimation functionality using population and exchange-rate data.
- Implemented database-backed caching to reduce repeated external API requests and provide locally accessible data.
- Containerised the application using Docker and fully documented the API and local development workflow.

AI Travel Planner Agent — AI-Powered Travel Planning Application

Python · REST APIs · AI Agent Workflows · External API Integration

- Developed an AI-powered travel planning application designed to generate practical international trip plans.
- Integrated external data sources to support travel planning, currency conversion, and budget estimation.
- Explored agent-based workflows for combining multiple data sources into structured travel recommendations.
- Deployed the application as an accessible web API and resolved API integration and request-handling challenges.

Data Encryption with Image & Audio Steganography — Secure Communication System

Java · AES Encryption · Image Steganography · Audio Steganography

- Designed and developed a Java-based secure communication system combining AES encryption with image and audio steganography.
- Implemented data encryption, embedding, extraction, and decryption workflows for secure information transfer.
- Conducted comparative performance analysis between image and audio steganography using processing time, accuracy, and file-size metrics.
- Applied principles of cryptography, information hiding, and secure data transmission in a practical software system.

EDUCATION

B.Tech, Computer Science — Second Class Upper

2017 – 2022

Abubakar Tafawa Balewa University (ATBU), Bauchi

CERTIFICATIONS & LEADERSHIP

CERTIFICATIONS	LEADERSHIP & ACHIEVEMENTS
• Huawei Certified ICT Associate (HCIA) – Datacom — Training Completed	• Auditor General — Mathematical Science Dept., ATBU
• Nigerian Computer Society (NCS) Certified	• Captain — NAMSSN Football Team
• Oracle Cloud Infrastructure (OCI) Foundations Associate	• Participant — Huawei ICT Competition
	• Two-time Best Football Player of the Year — NAMSSN
	• Member — Nigerian Computer Society (NCS)
	• Member — Huawei ICT Academy, ATBU